An extracellular-matrix gene program in clear-cell renal cell carcinoma stroma

Leviel Fluhr · Opeongo computational biology home assignment · 7 November 2026 · Code, environment and every number/figure: **github.com/FluhrLeviel/opeongo_assignment** (`uv sync && uv run opengo all`; decisions with evidence in `outputs/tables/decisions_log.tsv`)

Summary. (1) Of three RNA cohorts, cohort C was excluded because its counts are capped at ≤409 non-mitochondrial counts per cell; A (single-nucleus, 25 donors) and B (single-cell, 11 donors) were combined with Harmony (batch = cohort), which aligned cell identities without erasing tumor/adjacent structure. (2) The 10-gene panel is a stromal program; in the combined fibroblast + mural (pericyte) compartment, cohort B shows higher panel expression in tumor than adjacent tissue in **9/9 paired donors** (+1.04 SD, 95 % CI 0.63–1.46), robust to compartment composition. Cohort A (3 paired donors) is too imprecise to confirm or refute this, so the pooled estimate depends on B and on the pooling model. Fibrillar collagens, FN1 and SPARC drive the increase; DCN and LUM do not follow. (3) The 3 matched snRNA–bulk donors are too few to decide; the whole CPTAC bulk RNA (72 pairs) and protein (80 pairs) cohorts show the same gene-level pattern, which may reflect per-cell change, the expansion of mural and vascular tissue, or both.

1 · Quality, cohorts and integration

Loading and reconciliation. Cohort A raw droplet matrices were restricted to the 141,950 author-annotated nuclei (34 specimens; every annotation matched); B (157,881 cells, 26 specimens) and C (29,499 cells, 13 ccRCC/adjacent specimens; the chromophobe specimen is out of scope) were loaded in full. Per-specimen counts equal the package manifest. Genes were matched by Ensembl ID (A, C) and unique symbol (B; 532 symbols that are old and new names of the same gene were left out): 29,531 genes are shared, including all 10 panel genes. The suggested label guide maps A "TRM" to lymphoid; these 510 nuclei express CSF1R (50 %, as macrophages) but not CD3D/E, CD2 or CD8A (≤2 %), so they were remapped to myeloid.

QC criteria. Flags were set per specimen and per assay rather than with global cut-offs: counts/genes < median − 5 MAD, counts > median + 5 MAD, mitochondrial fraction > median + 3 MAD and >5 % in nuclei (where mitochondrial reads indicate ambient RNA) or >25 % in whole kidney cells, plus Scrublet doublets. A and B were already author-filtered (B's QC fields match our recomputation exactly); flagged cells were removed (A 4.2 %, B 0.9 %; no specimen removed). Removal is uneven across populations: ≤4 % in stroma and endothelium, but 22 % of A lymphoid, 10 % of A myeloid and 13 % of B non-malignant epithelial cells (Fig. 1f), one reason population proportions are never compared. **Cohort C was excluded**: every cell has ≤409 non-mitochondrial counts (A and B reach 10^4–10^5; Fig. 1a), 76 % of its reads are mitochondrial, its two worst specimens are the tumors of two of its four paired donors, and DCN/LUM are detected in 0.2 % of cells. A marker-based labelling validated on A and B (81 %/95 % agreement with author labels) still recovers broad populations in C, but per-cell ECM signal is too sparse.

Baseline vs integration. On a stratified subsample (87,489 cells: ≤1,000 per specimen plus every stromal and endothelial cell) the unintegrated A+B embedding separates by cohort (Fig. 1b). In whole-specimen pseudobulk PCA, PC1 carries 83 % of variance and is almost entirely cohort (R^2 0.99; single-nucleus vs single-cell). Cohort A alone, the starting point, has cell-type ASW 0.63 and tumor/adjacent structure within stroma 0.26. Harmony with batch = cohort (Fig. 1c,d) raised within-population cohort mixing (batch ASW 0.49→0.81), cell-type separation (0.62→0.65) and cross-cohort kNN label transfer (balanced accuracy 0.72→0.88), and kept the tumor/adjacent structure within stroma (0.24→0.20). Harmony with batch = specimen erased that structure (0.02), because specimen is confounded with condition, and was rejected (Fig. 1e). *Preserved:* all major populations (kNN purity 0.89–1.00) and within-population condition structure. *Weakened:* the malignant/non-malignant epithelial boundary across cohorts and global mixing, which stays low because the cohorts differ in composition (A mostly tumor nuclei, B mostly immune cells) and assay. *Lost:* nothing with batch = cohort; with batch = specimen, the tumor/adjacent structure. C's potential contribution (4 more paired donors) is lost to its exclusion. *What combining adds:* after Harmony, A's labels predict 99 % of B's fibroblast and pericyte cells as stroma (69 % unintegrated), supporting one stromal compartment across cohorts, and paired donors for that compartment rise from 3 to 12 (Fig. 1g). The embedding was used only for visualisation and label transfer; no test uses it.

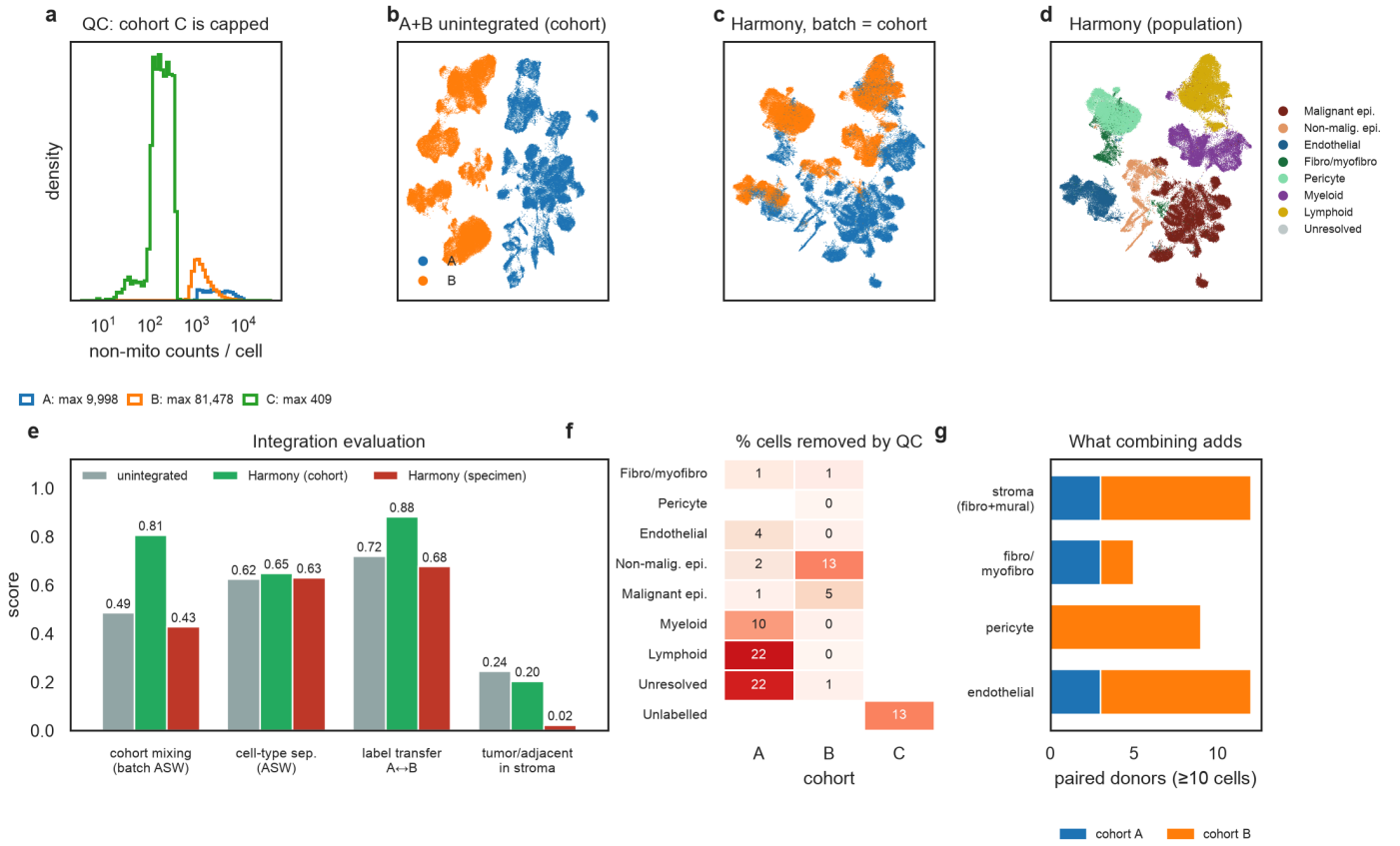


Figure 1. Quality control and integration. (a) Non-mitochondrial counts per cell: cohort C has a hard ceiling. (b–d) UMAPs of the A+B subsample: unintegrated and Harmony (batch = cohort), coloured by cohort or population. (e) Integration metrics: batch ASW = 1 − |silhouette of cohort| within each population; cell-type ASW = rescaled silhouette of populations; label transfer = mean A→B / B→A balanced accuracy of a 15-nearest-neighbour classifier; last group = tumor-vs-adjacent silhouette within stroma (mean of A and B). (f) Percentage of cells removed by QC per population. (g) Paired donors with ≥10 cells in both conditions per candidate population.

2 · The ECM program

Overview and summary. In both cohorts the stromal compartment has the highest expression of every panel gene (Fig. 2a); endothelial cells express SPARC and FN1 but few collagens and no DCN/LUM; immune and epithelial cells are low. Within stroma, DCN and LUM are fibroblast-specific (high in B fibroblasts; near zero in B pericytes and in A "fibroblasts", which by non-panel markers RGS5, NOTCH3 and PDGFRB are largely mural cells; Fig. 2a). As the panel summary we use the **mean of per-gene z-scored log2-CPM** (z within cohort and population): genes co-vary across stromal pseudobulks (median pairwise ρ 0.48 in A, 0.70 in B) and the score agrees with PC1 (ρ 1.00/0.97) and log summed CPM (0.96/0.98), so the simplest equal-weight score is used; DCN and LUM, which do not co-vary with the rest in B, are always reported separately.

Population choice. The **combined fibroblast + mural compartment** (A fibroblasts and myofibroblasts; B fibroblasts and pericytes) expresses the panel most strongly, is supported as one compartment by non-panel markers and label transfer, and has 3 + 9 paired donors with ≥10 cells. Fibroblasts alone have 5 pairs (2 in B); endothelium has 12 pairs but expresses mainly SPARC/FN1 and is used as a contrast. Malignant cells have no adjacent counterpart and were not tested.

Testing. Values tested are pseudobulk raw counts of QC-passing compartment cells per specimen, converted to log2-CPM with library size from all genes; specimens with <10 compartment cells are dropped. Donors are the replicates: repeated tumor regions are averaged per donor and each donor's tumor is compared with its own adjacent sample. Per cohort we report the mean paired difference, 95 % t-interval, direction count and an exact sign-flip p (all 2^n sign patterns; with n = 3 the smallest possible p is 0.25), with Benjamini–Hochberg correction over genes; cohorts are combined by fixed- and random-effects meta-analysis.

Stromal compartment, tumor – adjacent	Paired donors	Panel score (SD) [95 % CI]	Donors up	Sign-flip p
Cohort B (single-cell)	9	**+1.04 [+0.63, +1.46]**	9/9	0.004
Cohort A (single-nucleus)	3	+0.02 [−1.77, +1.81]	1/3	1.0
A + B, fixed effect / pooled sign-flip	12	+0.88 [+0.56, +1.21]	10/12	0.003
A + B, random effects (k = 2 cohorts)	12	+0.60 [−0.39, +1.59]		

Observations. In B, COL1A2, COL3A1, FN1 and SPARC rise in all 9 donors (log2FC +1.7 to +2.6, q = 0.01) and COL1A1, COL6A1 and COL6A2 also rise (q ≤ 0.03); **DCN and LUM do not** (−1.7 and −0.7, q ≥ 0.17) and vary by ±5 log2 units between donors (Fig. 2d), consistent with their source being the ~3 % of B's compartment that is author-labelled fibroblasts. A shows no consistent direction (Fig. 2c); a mixed model adding A's unpaired tumor-only donors (24 donors) gives −0.18 [−0.85, +0.48], but those donors add only between-donor information and the contrast still rests on 4 adjacent specimens. **Does adding cohorts change the finding?** A's interval contains B's estimate, so A is uninformative rather than contradictory; the pooled result depends on the model (the fixed-effect interval excludes zero, the random-effects one does not). The heterogeneity statistic (I^2 = 80 %) is unreliable with two cohorts, one of n = 3, each standardised by its own SD.

Robustness check (main uncertainty: compartment composition). If the B effect reflected more fibroblasts in tumor stroma, it should vanish in pure mural cells and without the fibroblast genes, and the fibroblast share should rise. It does not: pericytes only +1.27 [+0.85, +1.70], 9/9 up; panel without DCN/LUM +1.41 [+0.96, +1.86], 9/9 up (Fig. 2e); fibroblast share tumor − adjacent +0.04 [−0.06, +0.14], up in only 2/9 donors (Fig. 2f). The estimate is also stable with ≥50 cells (+1.07), one tumor region per donor (+1.04), a donor random-intercept model on all specimens (+1.11 [+0.85, +1.36]) and leave-one-donor-out (+0.93 to +1.15; Fig. 2e). Endothelial cells show no change (+0.13 [−0.26, +0.52]), so this is not a global shift in ECM transcripts.

Interpretation. In single-cell data, tumor-associated mural/pericyte cells carry more fibrillar collagen, FN1 and SPARC transcript than adjacent ones in the same patients. Higher transcript levels are not evidence of active remodelling, and whether the effect holds in single-nucleus data and in fibroblasts is not established: cohort A has 3 pairs, a different assay and a compartment of different make-up.

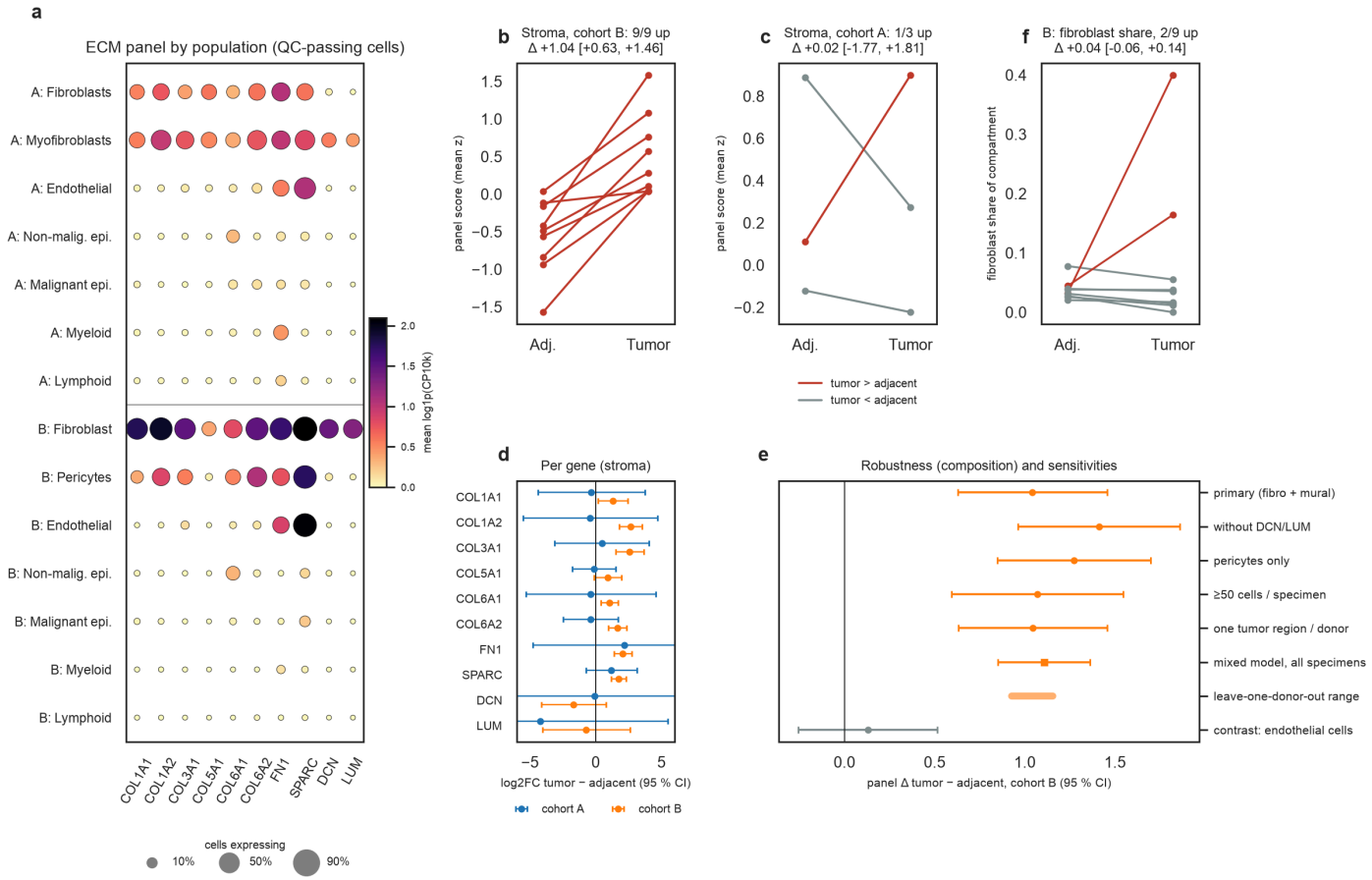


Figure 2. ECM panel across populations and tumor vs adjacent in stroma. (a) Fraction expressing (size) and mean expression (colour) per population (QC-passing cells). (b, c) Donor-level panel score in the fibroblast + mural compartment, adjacent vs tumor, cohort B and A. (d) Per-gene paired log2 fold change (95 % CI; A intervals clipped at ±6). (e) Cohort B robustness and sensitivity estimates (bar = leave-one-donor-out range; grey = endothelial contrast). (f) Fibroblast share of B's compartment per donor.

3 · Linked bulk RNA and protein

Correspondence. Every cohort A snRNA specimen was checked through the crosswalk: 29/34 are linked, all resolve through `bulk_samples.tsv` to real columns of the condition-specific RNA and protein files, and all protein aliquots match the author sample table. The two parent-specimen mismatches are exactly the two flagged links (C3L-00908, different parent; C3N-00242, protein from two parents), leaving 27 primary links (23 tumor, 4 adjacent). Donors with primary-linked tumor *and* adjacent are **C3L-00079, C3L-00088 and C3N-01200 (n = 3)**; C3N-00242 is a sensitivity case. Unlinked specimens were not paired by donor identity (per-specimen record: `stage6_specimen_inclusion.tsv`). Each assay used a different portion of the parent specimen, and bulk RNA is a donor/condition column whose exact aliquot is not reconciled. Bulk values were used as supplied (log2, normalised); panel genes have no missing values, although upstream imputation cannot be ruled out.

Within-assay comparison. Observations. In the matched donors (Fig. 3a,d) the panel difference is −0.22 (snRNA stroma; linked tumor specimen only, hence not the +0.02 of Section 2, which averages tumor regions), −0.14 (snRNA whole specimen), +0.48 (bulk RNA) and +0.26 (protein). Donors disagree and all p ≥ 0.5, so the matched subset cannot confirm or refute the finding; on average FN1 and SPARC are higher in tumor in every matched assay. As context, the whole CPTAC cohorts (the same study as cohort A, including the matched donors, mostly without snRNA) show higher panel levels in tumor in bulk RNA (+0.52 SD [0.31, 0.73], 54/72 donors) and protein (+0.68 [0.49, 0.87], 67/80), with **the same gene pattern as cohort B**: collagens, FN1 (RNA 72/72) and SPARC (RNA 72/72, protein 78/80) up; DCN down (RNA 4/72 up) and LUM down or unchanged. COL6A1 is the one discrepancy (down in bulk RNA, up in protein and B).

Composition. Non-panel marker genes in bulk RNA show that tumors lose fibroblast markers (PDGFRA, C7, FBLN1; 97 % of donors, −1.54 SD) and gain mural (+0.56) and endothelial (+1.63) markers. The 8-gene ECM change (without DCN/LUM) tracks fibroblast-marker (ρ 0.68) and mural-marker (ρ 0.62) change across donors (Fig. 3b,c) and rises on average (+0.96 SD) despite the fibroblast loss. **Interpretation.** Composition strongly shapes bulk differences. The rise is not explained by more fibroblasts, but it may reflect expansion of mural and endothelial tissue as well as, or instead of, per-cell change; bulk data cannot separate the two.

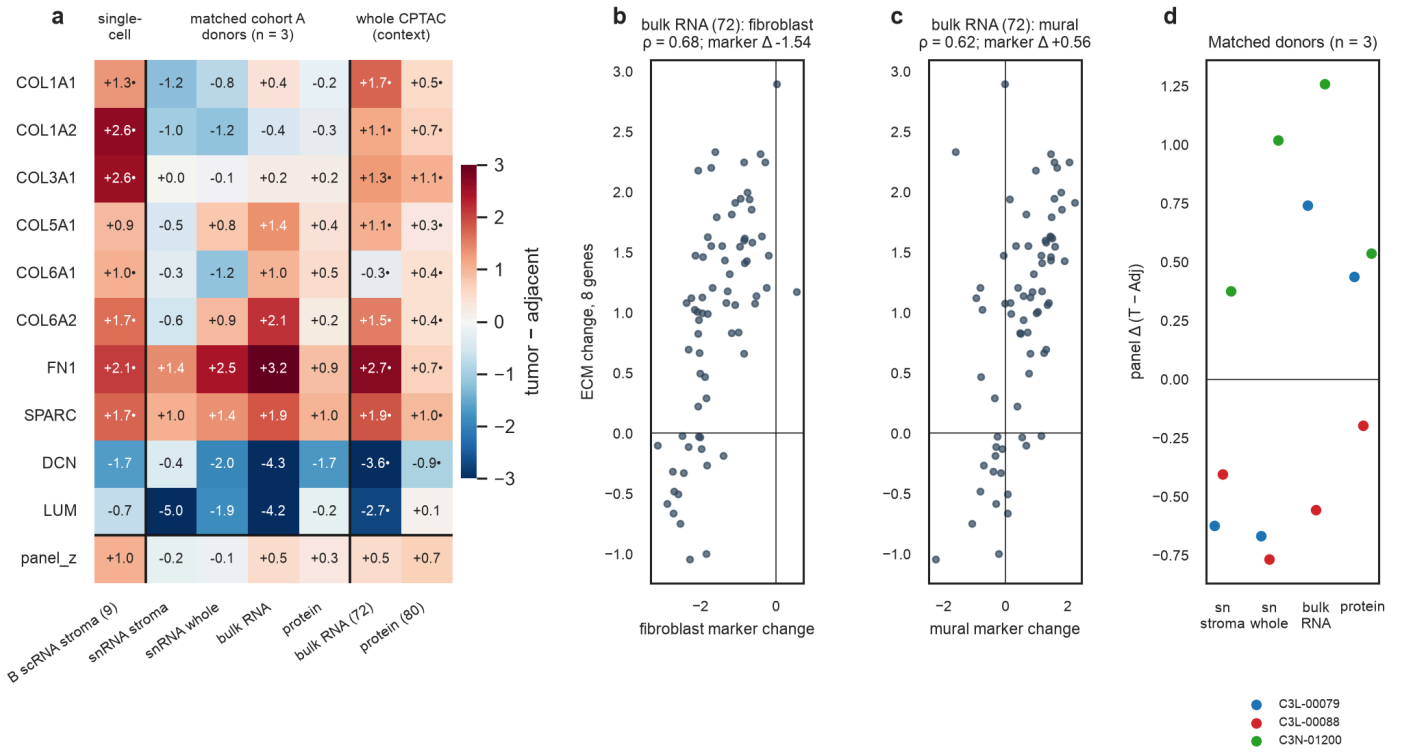


Figure 3. Linked modalities. (a) Tumor – adjacent per gene and panel, each estimated within its own assay: cohort B stroma (Section 2), the 3 matched cohort A donors (four assays), and the whole CPTAC bulk RNA/protein cohorts (context, not matched); dots mark BH q < 0.05. (b, c) Bulk RNA: 8-gene ECM change vs fibroblast (b) and mural (c) marker change per donor. (d) Panel difference per matched donor in each assay.

Agreement, discrepancies and limitations

- **Agreement:** single-cell B, bulk RNA and protein agree that collagen I/III, FN1 and SPARC are higher in ccRCC than in adjacent kidney, while DCN/LUM are not. **Discrepancy:** cohort A (single-nucleus) does not reproduce the stromal increase, and COL6A1 differs between bulk RNA and protein.
- **Cellular source:** cell-specific RNA does not establish the source of bulk protein. B places part of the RNA increase in pericytes, but endothelial cells, which expand in tumors, also express FN1 and SPARC strongly.
- **Coverage:** 4 adjacent specimens in A (3 usable pairs; smallest possible p = 0.25); 3 matched multimodal pairs; ~97 % of B's compartment is author-labelled pericytes, so "stroma" differs between cohorts.
- **Composition and capture:** single-nucleus and single-cell assays capture different cells; bulk differences are strongly composition-dependent; the marker-based composition proxy is approximate; ambient RNA in nuclei may add low-level ECM signal to non-stromal populations.
- **Exclusions:** cohort C (integrity), QC-flagged cells, specimens with <10 compartment cells, two flagged crosswalk links (sensitivity only), and five unlinked snRNA specimens for Section 3 (per-specimen records: `stage5_specimen_inclusion.tsv`, `stage6_specimen_inclusion.tsv`).

Abbreviations. ASW, average silhouette width; kNN, k-nearest neighbours; MAD, median absolute deviation; CPM, counts per million; log2FC, log2 fold change; SD units, per-gene SD of log2-CPM across the cohort's specimens; q, Benjamini–Hochberg-adjusted p; I^2, share of variance between cohorts; mural, pericyte/vascular smooth-muscle-like cells; CPTAC, Clinical Proteomic Tumor Analysis Consortium.

1. Heumos L. et al. Best practices for single-cell analysis across modalities. *Nat Rev Genet* 2023.
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7. Data: GEO GSE227898 (A), GSE178481 (B), GSE159115 (C); CPTAC ccRCC processed RNA and harmonised protein (see `SOURCES.md`).